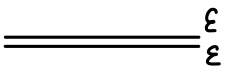
④ N non-interacting, distinguishable particles — microcanonical system (ensemble)

a) N, V, E constant



total energy is E

from N number of N_1 particle can be found on energy level ε

so: $E = N_1 \cdot \varepsilon$

microstate: from N_1 a particle can be on two levels, so the number of possible states: 2^{N_1}

How many ways can we choose N_1 from N if the particles are distinguishable?

$$\frac{N\cdot(N-1)\cdot(N-2)\cdot\ldots\cdot(N-N_1+1)}{N_1!} = \binom{N}{N_1}$$

$N_1!$ → ordering

$$\Omega(N,E) = \binom{N}{N_1}\cdot 2^{N_1} = \binom{N}{E/\varepsilon}\cdot 2^{\frac{E}{\varepsilon}} \quad \text{(independent particles)}$$

b) $$S(N,E) = k_B \ln \Omega(N,E) = k_B \ln\left[\binom{N}{N_1}\cdot 2^{N_1}\right] =$$

$$= k_B \cdot \ln\left(\frac{N!}{N_1!(N-N_1)!}\cdot 2^{N_1}\right) = k_B N_1 \cdot \ln 2 + k_B\cdot\left[\ln N! - \ln N_1! - \ln(N-N_1)!\right]$$

Stirling-formula: $\ln N! \approx N\cdot\ln N - N$

$$S(N,E) = k_B N_1 \cdot \ln 2 + k_B\left(N\cdot\ln N - N - N_1\cdot\ln N_1 + N_1 - (N-N_1)\ln(N-N_1) + N - N_1\right) =$$

$$= k_B\left[\ln 2\cdot\frac{E}{\varepsilon} + N\cdot\ln N - \frac{E}{\varepsilon}\ln\frac{E}{\varepsilon} - \left(N-\frac{E}{\varepsilon}\right)\ln\left(N-\frac{E}{\varepsilon}\right)\right] =$$

$$= k_B\cdot\left[N\cdot\ln N + \frac{E}{\varepsilon}\cdot\ln\frac{2\varepsilon}{E} - \frac{N\varepsilon - E}{\varepsilon}\ln\frac{N\varepsilon - E}{\varepsilon}\right] =$$

$$= k_B\cdot\left[N\cdot\ln N + \frac{E}{\varepsilon}\cdot\ln\frac{2\varepsilon}{E} - N\ln\frac{N\varepsilon - E}{\varepsilon} + \frac{E}{\varepsilon}\cdot\ln\frac{N\varepsilon - E}{\varepsilon}\right] =$$

$$= k_B \left[N \cdot \ln\left(N \cdot \frac{\varepsilon}{N\varepsilon - E} \right) + \frac{E}{\varepsilon} \ln\left(\frac{2\varepsilon}{E} \cdot \frac{N\varepsilon - E}{\varepsilon} \right) \right] =$$

$$= k_B N \left[\ln \frac{N\varepsilon}{N\varepsilon - E} + \frac{E}{N\varepsilon} \ln \frac{2(N\varepsilon - E)}{E} \right]$$

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c) Definition:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_N = k_B N \cdot \left[\frac{N\varepsilon - E}{N\varepsilon} \cdot \frac{N\varepsilon}{(N\varepsilon - E)^2} + \frac{1}{N\varepsilon} \ln \frac{2(N\varepsilon - E)}{E} + \right.$$

$S = S(E; N)$ — variables

$$\left. + \frac{E}{N\varepsilon} \cdot \frac{E}{2(N\varepsilon - E)} \cdot \frac{-2E - 2N\varepsilon + 2E}{E^2} \right] = \frac{k_B}{\varepsilon} \cdot \ln \frac{2(N\varepsilon - E)}{E}$$

change in E

$$T = \frac{\varepsilon}{k_B} \cdot \frac{1}{\ln \frac{2(N\varepsilon - E)}{E}}$$

Two system in thermal contact, at equilibrium maximum entropy:

$\frac{\partial (\ln \Omega)}{\partial E} = 0$ $\quad \ln \Omega = \ln \Omega_1 + \ln \Omega_2$

$\Rightarrow \frac{\partial \ln \Omega_1}{\partial E_1} = \frac{\partial \ln \Omega_2}{\partial E_2}$

$\Rightarrow$ temperature

if $E \geq \frac{2N\varepsilon}{3}$: $\frac{N\varepsilon}{E} = x \Rightarrow x \leq \frac{3}{2}$

$$T = \frac{\varepsilon}{k_B} \cdot \frac{1}{\ln(2x-2)} \Rightarrow 2x - 2 \leq 1$$

$$\ln(2x-2) \leq 0 \Rightarrow \begin{cases} x = \frac{3}{2} : T \to 0 \\ 1 < x < \frac{3}{2} : T \leq 0 \end{cases}$$

$\Rightarrow$ microcanonical ensemble cannot describe the system

increasing E means decreasing S

d) $C_V = \left(\frac{\partial E}{\partial T} \right)_V$; $\quad \frac{\varepsilon}{k_B T} = \ln\left[2\left(\frac{N\varepsilon}{E} - 1 \right) \right] \Rightarrow E = \frac{2N\varepsilon}{e^{\frac{\varepsilon}{k_B T}} + 2}$

$$C_V = \frac{2N\varepsilon}{\left(e^{\frac{\varepsilon}{k_B T}} + 2 \right)^2} e^{\frac{\varepsilon}{k_B T}} \cdot \frac{\varepsilon}{k_B T^2}$$

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e) $dS = \left(\frac{\partial S}{\partial E} \right)_N dE = \left(\frac{\partial S}{\partial E} \right)_N \cdot \left(\frac{\partial E}{\partial T} \right)_V \cdot dT = \frac{dT}{T} \cdot C_V = \frac{\delta Q}{T}$